Question ID: 58443765

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$$y = 5x + 4$$

$$y = 5x^2 + 4$$

Which ordered pair (x, y) is a solution to the given system of equations?

Answer

A. $(0, 0)$

B. $(0, 4)$

C. $(8, 44)$

D. $(8, 84)$

Correct Answer: B

Rationale

Choice B is correct. The second equation in the given system is $y = 5x^2 + 4$. Substituting $5x^2 + 4$ for y in the first equation of the given system yields $5x^2 + 4 = 5x + 4$. Subtracting 4 from both sides of this equation yields $5x^2 = 5x$. Subtracting $5x$ from both sides of this equation yields $5x^2 - 5x = 0$. Factoring out a common factor of $5x$ from the left-hand side of this equation yields $5x(x - 1) = 0$. It follows that $5x = 0$ or $x - 1 = 0$. Dividing both sides of the equation $5x = 0$ by 5 yields $x = 0$. Substituting 0 for x in the equation $y = 5x + 4$ yields $y = 5(0) + 4$, or $y = 4$. Therefore, a solution to the given system of equations is the ordered pair $(0, 4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.